

Fig. 11: Channel 3 settings for load 1 to turn on/off through RL1

PARTS LIST

Semiconductors:

Board1	- ESP8266 NodeMCU (ESP-12)
IC1	- 7805, 5V voltage regulator
IC2-IC3	- 4N33 opto-coupler
D1-D2	- 1N4007, rectifier diode
LED1-LED3	- 5mm LED
T1-T2	- BC547, npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 3.3-kilo-ohm
R2, R5-R8	- 1-kilo-ohm
R3	- 10-kilo-ohm
R4	- 4.7-kilo-ohm

Capacitors:

C1	- 100 μ F, 25V electrolytic
C2	- 0.1 μ F ceramic disc

Miscellaneous:

DHT11	- Temp. and humidity sensor
S1	- On/off switch
CON1	- 4-pin connector
BATT.1	- 12V battery
RL1-RL2	- 12V, single-changeover relay
LDR1	- 5mm LDR
	- 7-pin Berg connector (female)
	- 2-pin terminal connector for battery



Fig. 12: Final dashboard of the project on Cayenne IoT platform

right. You will get a screen as shown in Fig. 7.

Click on plus (+) sign on each channel icon to add the channel to your dashboard. You will get the names of added channels on the left under Add New page as shown in Fig. 8.

Click on Settings of Channel 0 and change its Widget name to, say, Luminosity. Click on Choose Icon, select Luminosity and click on Save. The icon will change as shown in Fig. 10. This will show the light around LDR in percentage.

Similarly, change the name of Channel 1 as Temperature and Channel 2 as Humidity. Make the required settings as mentioned in Channel 0. Now, proceed to channels 3 and 4 for configuring the buttons for controlling

the two relays. Go to Settings of channels 3 and 4 and delete these using Remove button.

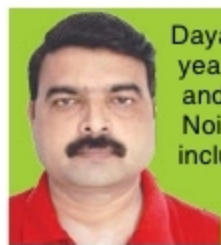
Now, click on Add New→Device→Widget. Select Custom Widgets→Button. Fill the required data for load 1 as shown in Fig. 11. Click on Add Widget. Similarly, repeat the load settings for Channel 4.

The final dashboard is shown in Fig. 12. Now, you are ready to control the two relays attached to channels 3 and 4. You can also monitor luminosity, temperature and humidity level of a place where your devices are located.

Download Cayenne app from Google Play Store, and login using the same credentials to start controlling the above-mentioned parameters from anywhere over the Internet. **EFY**

EFY Note

The source code of this project is available for free download at source.efymag.com



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